Group Members:

Muhammad Ali Haris 22k-4239

Muhammad Abdullah Shariq 22k-4497

Hamza Haroon 22k-4200

Project Proposal: Matrix Multiplication using OpenMP

1. Project Title

Parallel Matrix Multiplication using OpenMP

2. Objective

The objective of this project is to implement and analyze matrix multiplication using parallel programming techniques with OpenMP. The project aims to improve computational efficiency by utilizing multiple threads and comparing the performance with the traditional sequential approach.

3. Problem Statement

Matrix multiplication is a fundamental operation in computer science and is widely used in applications such as graphics, machine learning, and scientific computing. However, for large matrices, the sequential implementation becomes computationally expensive and timeconsuming. This project addresses this issue by parallelizing the matrix multiplication process using OpenMP to achieve faster execution.

4. Proposed Solution

The project will implement matrix multiplication using OpenMP in C/C++. The computation will be parallelized using OpenMP directives such as `#pragma omp parallel for`.

Key Features:

- Generate two matrices of size N × N
- Perform matrix multiplication sequentially
- Perform matrix multiplication using OpenMP
- Measure execution time of both approaches
- Compare performance (speedup and efficiency)

5. Methodology

Step 1: Sequential Implementation

- Implement standard matrix multiplication using nested loops.

Step 2: Parallel Implementation

- Use OpenMP directives to parallelize loops.
- Distribute rows of the matrix across threads.

Step 3: Performance Analysis

- Measure execution time using `omp_get_wtime()`
- Calculate speedup:

  Speedup = Sequential Time / Parallel Time

Step 4: Experimentation

- Test with different matrix sizes (e.g., 500×500, 1000×1000)
- Test with different numbers of threads

6. Tools and Technologies

- Programming Language: C/C++
- Parallel Framework: OpenMP
- Compiler: GCC with OpenMP support (`-fopenmp`)

- Platform: Linux / Windows (WSL)

7. Expected Outcomes

- Reduced execution time using parallel processing
- Demonstration of performance improvement
- Graphs showing comparison between sequential and parallel execution

8. Applications

- Scientific computing
- Machine learning algorithms
- Computer graphics
- Data processing systems

9. Conclusion

This project will demonstrate how parallel computing can significantly improve performance in computational tasks like matrix multiplication. It will provide practical understanding of OpenMP and thread-based parallelism.